
Categories: the definition

We are finally ready for the formal definition of a category, but we'll discuss each part of the definition first.

We're going to define a framework for thinking about relationships. In order to make sure we are thinking about *slightly* reasonable types of relationships, we will stipulate that they at least behave slightly well, so that we can do various things with them. These stipulations will be inspired by some of the things we've already seen, but will be less stringent than the stipulations on equivalence relations. We will talk through all the parts of the definition first, and then present it formally in Section 8.4.

We are building up to the definition of a category, and we'll call it $\mathcal{C}$ (a curly C). Just like when we talk about person A or a number x this is not a particular category, but a general category that we have named with a letter for convenience.

Our definition takes a typical form for abstract mathematical structures, in three parts:

- Data: basic building blocks.
- Structure: things we can do with those building blocks.
- Properties: rules about how those things must fit together.

8.1 Data: objects and relationships

Our basic data in a category are objects and relationships. The objects can be anything, we just have to be clear what we're including. This is a bit like when we defined relations starting with the “world” S of objects we were going to consider. For a category $\mathcal{C}$ we might call the world of objects $\text{ob } \mathcal{C}$. In a particular category these could be numbers, people, shapes, mathematical structures of a particular kind, points in a space — really anything. They could also be just unspecified abstract objects. We'll look at plenty of examples soon.

The relationships might be things like “less than” or “is the mother of”, but

they also might not be anything like relationships, so we more generally call them arrows, or “morphisms”, inspired by examples where they transform or “morph” one thing into another. (However in this book we might still informally refer to them as relationships). We draw them like this: $a \longrightarrow b$, as we have done for many relationships already in this book. As the notation indicates, each arrow in a category starts at an object and ends at an object. The beginning object is called the “source” and the end object is called the “target”. Sometimes these are called domain and codomain, but personally I find source and target much more vivid; in the spur of the moment I can’t actually remember which is which out of the domain and codomain.

Unlike the relations we looked at in the previous chapter, the ones in a category aren’t just true or false. Given two objects a and b , we’re not just asking if they’re related or not; we ask a more nuanced question (as we should in life as well) which is: *in what ways* are these objects related?

This means that there could be more than one arrow from a to b and we need to give them names, perhaps like this.

$$a \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} b$$

When there’s more than one arrow with the same source and target, as above, we call them *parallel*, because they look parallel.

Another way to have multiple arrows between the same objects is pointing in opposite directions like this.

$$a \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{g} \end{array} b$$

There is no limit to the number of arrows going from any object to another. There could be infinitely many and there could be none.

We often focus on one pair of objects a and b and write the collection of arrows from a to b as $\mathcal{C}(a, b)$. Some people[†] write it as $\text{hom}_{\mathcal{C}}(a, b)$, which comes from the idea of “homomorphisms” which we’ll see quite a bit of later on. The set of morphisms $\mathcal{C}(a, b)$ is sometimes called a “homset”. Homsets are directional: if an arrow starts at b and ends at a then it will be in $\mathcal{C}(b, a)$ and not in $\mathcal{C}(a, b)$.

8.2 Structure: things we can do with the data

Next we declare that there are a couple of things we can definitely do with these relationships. The first is a bit like reflexivity from the previous chapter: we want to make sure every object is related to itself. Note that we haven’t stipulated a particular type of relationship so this can be quite a weak condition. We’re not saying “everything is less than itself” or “everyone is a friend of

[†] I’m mostly too lazy to write all those extra characters.

themselves” we’re just observing that everything *is the same as* itself and that this most basic relationship should be encapsulated in every category.

So we say that given any object a there must be at least an arrow like this, called an *identity*.

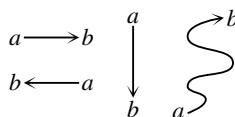
$$a \xrightarrow{1_a} a$$

We often write it as 1_a , hence the label. We could also draw it going in a little loop, like this.

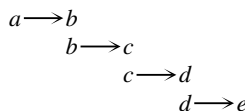


If we draw it the first way, as a straight line, we’re not saying that there are two objects called a , we’re specifying that the source and target of this particular arrow are both a . Incidentally, when we draw arrows their shape doesn’t make any difference to what they are. In a category, the thing that matters is the endpoints of the arrow, not the physical path the arrow takes in between them.

We could draw this arrow in any of these ways and it would be logically the same information, although sometimes it helps our geometrical intuition to draw it different ways up in different situations.



The other thing we want to be able to do with arrows is a bit like transitivity: we want to be able to build up strings of relationships. If we draw this analogously to how we did in the previous chapter it might now come out as shown on the right.



However it’s a bit tedious to keep writing all those repeated letters, so we could draw it like this.

$$a \longrightarrow b \longrightarrow c \longrightarrow d \longrightarrow e$$

Then we want to get out an arrow like this.

$$a \longrightarrow e$$

But as with transitivity, we only have to specify this for two arrows at a time, the idea being that we can then gradually build up to long strings (subject to some conditions we’ll think about in a moment).

So what we actually say is that given two arrows f and g in this “nose-to-tail” configuration

$$a \xrightarrow{f} b \xrightarrow{g} c$$

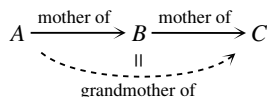
we want to be able to combine them into a single arrow like this:

$$a \longrightarrow c$$

This process of combining arrows is called *composition*. The answer we get out at the end is a new arrow called “ f composed with g ” and we write it as $g \circ f$ which is a bit annoying because it looks like we’ve written it backwards. We sort of have, and some people prefer writing it the other way, but the reason I (and many other mathematicians) prefer it this way is that it makes sense for composition of functions. We’ll come back to that in the next chapter.

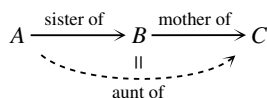
This is similar to transitivity, but different. The relations of the previous chapter were yes/no situations: things just were or weren't related. Now we're in "sense in which" situations, so this is more like saying: if there's a sense in which a is related to b and a sense in which b is related to c then we can combine those to get a sense in which a is related to c . If we think of it as paths then it's like saying "if there's a path from a to b and a path from b to c we can do one and then the other to get a path from a to c ".

Example 8.1 We have seen that the relation "is the mother of" is not transitive. But we can *compose* it, and draw the process like this.



I've drawn the composite as a dotted arrow to remind us that this is the one we added afterwards, deducing it from the others. This is where videos really help with category theory, because the diagrams are often "dynamic" in that you draw part of the diagram first and then you deduce some other part.

Here's another example involving some different family relationships; see if you can practice reading it "dynamically".



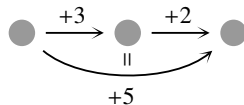
I hope you can now see that the diagrams in Chapter 5 were all secretly diagrams of categorical composition. ("Categorical" means something else in normal life, but in category theory we use it to mean "category theoretical".)

These examples show that composition of arrows is more general than transitivity as it includes examples that transitivity excluded, like "is the mother of". But furthermore, we can apply it to situations where the transitivity question couldn't even be *posed*, like the example producing an aunt.

Composition is also more general than the notion of a binary operation. It does appear to take two inputs (the two arrows we're composing) and produce one output (the composite), but we can't just take *any old* two arrows and produce a composite: they have to be "composable" arrows, that is, where the target of the first one matches the source of the second one. It's a bit like the question of connecting flights, when it appears that you can fly somewhere by changing plane in the middle, but when you click on the details it admits that you also have to switch airport, say from London Heathrow to London Gatwick (which is really quite arduous). We don't want that kind of composite: we want the two paths to *really* meet in the middle.

This makes it a generalization of binary operation. Perhaps it's easier to see that binary operations are a special case of composition, just with nothing really happening at the endpoints. That is, in the special case that all the endpoints are the same, all arrows can be composed so it's just a binary operation.

This is like in Section 5.4 when we saw diagrams like this to think about addition. We were essentially representing numbers as arrows, and composition was the binary operation $+$.



We'll come back to these ideas when we talk about monoids in the Interlude; the main thing to take away now is that composition of arrows is a generalization of the concept of binary operation.

Note on grammar

Note that composition comes with various parts of speech:

- “to compose” (verb): we compose arrows together
 - “composition” (noun): the act of composing
 - “composite” (noun): the result of composing arrows together is a new arrow called the composite, or composite arrow.
-

Things To Think About

T 8.1 We introduced the idea of composition with the idea of building up long strings of arrows, but composition only ever refers to two arrows. When we talked about transitivity we saw that we could use two instances of $\square$ to build strings of any length. Can you see why this is harder for composition of arrows? We need another condition, which we'll address in the next section.

8.3 Properties: stipulations on the structure

So far we have our data and our structure. Now we are going to impose some conditions on the structure, the “properties” that will ensure our structure behaves in a most basically sensible way. There is one condition for identities and one for composing more than two arrows. Properties are more formally called *axioms*. They are basic laws governing our structure.

Identities

We said that identities just encapsulate the fact that everything is the same as itself. However so far all we have said is that the source and target are the same. The property we need is a bit like other situations where an element[†]

[†] By “element” I just mean “thing” here.

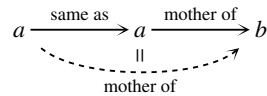
doesn't *do* anything, and those elements are generally called identities. If I line up these conditions I hope the analogy is clear.

0 is the identity for + on (real) numbers: $\forall a \in \mathbb{R} \quad a + 0 = a$

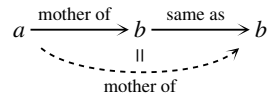
1 is the identity for $\times$ on (real) numbers: $\forall a \in \mathbb{R} \quad a \times 1 = a$

Now instead of addition or multiplication our identity is going to “do nothing” with respect to composition of arrows.

It's a bit like this, where sticking “same as” at the beginning doesn't give us any extra information about the relationship between a and b .



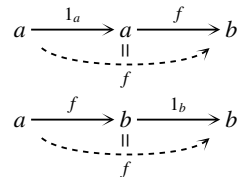
We could stick “same as” at the other end as well, and it still won't make any difference.



Things To Think About

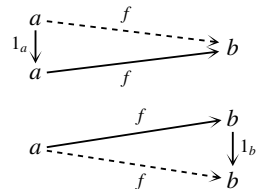
T 8.2 See if you can write down the general condition on identity arrows.

For general identity arrows we take the above blueprint but replace “mother of” with a general arrow $a \xrightarrow{f} b$, and replace “same as” with the appropriate identity arrow, giving these two conditions sometimes called *unit laws*.



In the first case, the only thing we know in advance is that the composite is $f \circ 1_a$, and in the second case the composite is $1_b \circ f$. So what we are stipulating here is $f \circ 1_a = f = 1_b \circ f$ which shows the analogy with the additive and multiplicative identities from earlier. We have to specify what happens with the identity on either side of f here, because unlike with $+$ and $\times$ there isn't any commutativity to rely on.

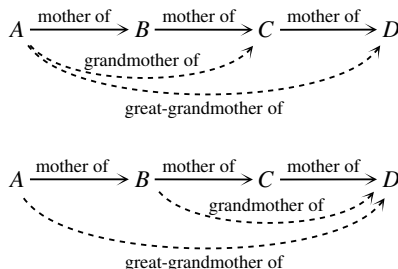
The expression as a string of symbols makes it look more like the additive and multiplicative identities from before, but drawing arrows helps us see what's going on better, I think. It might be even more visually vivid if I change the geometry a bit, without changing the logical structure.



This flexibility of layout is a powerful feature of arrow diagrams, as adjusting the geometry can help us invoke different kinds of intuition.

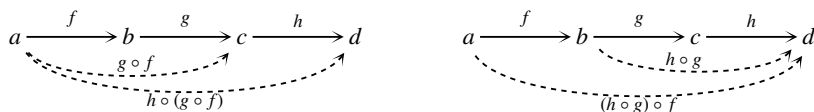
Composing more than two arrows in a string

Let's go back to family relationships and consider three generations of mothers. There are now two ways we could compose these arrows. We could go from A to C and then include D , or we could go from B to D and then include A , as shown in these two diagrams.



In this situation both ways produce the same relationship between A and D . This is *associativity* and we will now state it in the general situation.

Suppose we have a string of three composable arrows: $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$. Then we have the following two ways of composing them:



We want these to be the same, that is, we want the equation on the right to hold.

$$h \circ (g \circ f) = (h \circ g) \circ f$$

This string of symbols makes it look closely analogous to associativity of binary operations like addition and multiplication.

$$h + (g + f) = (h + g) + f$$

However it's a little more subtle for arrows because f, g and h have to be *composable* arrows, not just any old arrows. The algebraic equation makes it look familiar, but the arrow diagram gives us a more subtle and also richer point of view as we'll see shortly.

Note that associativity looks like it's a condition on three composable arrows, and it is, but what we really want to know is that a string of composable arrows of *any* length has one unambiguous composite.[†] This is a small difference in perspective at this level, but it gets magnified into something quite serious in higher dimensions. There is an interplay between all the things you want to be true in the world you're studying, and finding a small tractable set of conditions you can check, that will ensure all those things are true. There is then a choice between what you take as the definition and what you take as a theorem. For associativity the choice is this:

[†] If you know about proof by induction, you could try proving this by induction on the length of the string; probably the hardest part is writing down precisely what we're trying to prove.

	Approach 1	Approach 2
definition:	strings of length 3	strings of length n
theorem:	strings of length n follow	strings of length 3 suffice

In higher levels of math I would say the second type of approach becomes more prevalent. Unfortunately at lower levels the first is more prevalent, which can make things seem rather arbitrary, because the “small tractable set of conditions” that make something broad true can obscure the nature of the broad thing that we really want. It’s like the broad thing is the wood (forest) and the tractable set is the trees, and if we focus on the trees we might miss the forest.

8.4 The formal definition

We have talked our way through the definition of category but now we’d better write out the complete definition formally. Drumroll. . .

A *category* $\mathcal{C}$ consists of:

DATA

- Objects: a collection $\text{ob } \mathcal{C}$ of objects
- Arrows: for objects $a, b \in \text{ob } \mathcal{C}$, a collection $\mathcal{C}(a, b)$ of arrows $a \longrightarrow b$

STRUCTURE

- Identities: for any object $a \in \text{ob } \mathcal{C}$ an identity arrow $a \xrightarrow{1_a} a$
- Composition: for any arrows $a \xrightarrow{f} b \xrightarrow{g} c$, a composite arrow $a \xrightarrow{g \circ f} c$

PROPERTIES

- Unit laws: for any arrow $a \xrightarrow{f} b$, $f \circ 1_a = f = 1_b \circ f$
- Associativity: for any arrows $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$, $(h \circ g) \circ f = h \circ (g \circ f)$

Arrows are also called morphisms, and the composite $g \circ f$ is also written gf .

What happened to symmetry?

We talked about how this definition is inspired by equivalence relations, with identities being like reflexivity and composition being like transitivity, so you might be wondering what happened to symmetry. The answer is that we don’t *demand* something analogous to symmetry in a category as we want to allow for the possibility of relations that only go one way. Instead of *demanding* this, we will *look for* it. This will be the notion of invertibility, a more subtle notion of sameness in categories, and we will get to that in Chapter 14.

8.5 Size issues

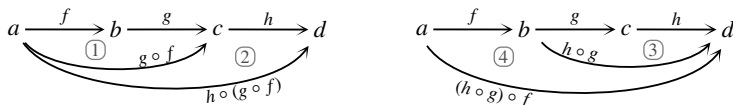
There is a technical issue to do with “size” that we need to be aware of, but we don’t need to make a big meal out of it. We just have to make sure we avoid the sort of paradox which arises if you attempt to think about a “set of all sets” or something similarly self-referential. This is the essence of Russell’s Paradox, and I’ve put a few more details in Appendix C in case you want some more explanation. Basically, to avoid getting into a paradox in the foundations of mathematics we have to put a hierarchy of size on collections of things, and in particular the collection of all sets doesn’t count as a set: it’s a “collection” and is one level up in the hierarchy. This then gives us a concept of different sizes of category, defined as follows:

- If the objects and the morphisms each form a set the category is called *small*.
- If each individual collection $\mathcal{C}(a, b)$ is a set then the category is called *locally small*; in this case the objects might form a collection, not a set.

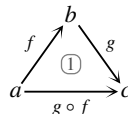
This becomes more important when one gets into more technical details, but for now if it baffles you I think it’s fine to just move on and think about it later if you need to, or have a look in the Appendix if you’re interested.

8.6 The geometry of associativity

In the explanation of identities I included a slightly different diagram that brought out some more of the geometry of the situation. I’d like to do the same for associativity now, but it’s a bit harder as it’s really going to be in three dimensions. The idea to keep in mind all the way through is that with the arrow notation it doesn’t matter what physical shape the arrows take, it only matters what the endpoints are. Let’s think about the arrow diagrams for associativity again, and this time I’ve marked in the four regions.



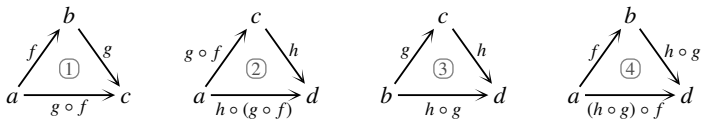
Region 1 has three arrows bordering it, and if we redraw them as straight lines and change the positions a bit, we can see it as a triangle.



Things To Think About

T 8.3 Try and draw out the other three regions as triangles, and label the corners and arrows. If you can, cut them out as actual triangles and then see if you can fit them together as a sort of 3-D jigsaw. Don’t forget to use associativity.

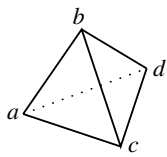
The four triangles look like this:



We can then fit them together in a 3-dimensional shape as long as we observe:

- By associativity we know $(h \circ g) \circ f = h \circ (g \circ f)$ so those two edges are allowed to go together.
- Some of the triangles as drawn will face in and some will face out.

This is pretty hard to describe on a 2-dimensional page so I encourage you to cut out some triangles and fiddle around with it a bit. The shape it makes is a tetrahedron, that is, a pyramid with a triangular base, a bit like this.



The geometry of associativity is something that gets studied a lot when we think more about higher dimensions.

Things To Think About

T 8.4 We have seen a progression through dimensions as in this table.

What happens if we try and continue this table up to more dimensions? So at the next level we take 5 objects, with 4 composable arrows, and see what shape we make.	no. of objects	no. of composable arrows	no. of dims	shape
	1	0	0	point
	2	1	1	line
	3	2	2	triangle
	4	3	3	tetrahedron

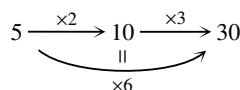
8.7 Drawing helpful diagrams

The diagrams we draw in category theory aren't exactly part of the definition — you can have objects and arrows whether you physically draw them or not. However, drawing diagrams can help by enabling us to invoke our geometric intuition. When we draw the diagrams there are various principles we use to draw helpful, rather than messy, diagrams.

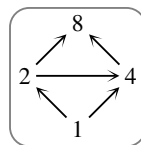
1. The physical positioning of the diagram doesn't affect the logical structure, so we can move things around as long as the arrow connections stay the same. This is key to making geometrically helpful diagrams.

2. We usually don't bother drawing identity arrows because they're always there. Occasionally we draw them if we're trying to emphasize something, for example if two arrows compose to the identity.
3. We omit arrows whose existence we can deduce from composition.

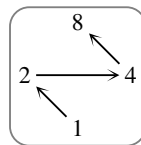
We saw this last point in Chapter 5 where we observed that we don't need to draw relationships that can be deduced, such as in factor diagrams.



This means, for example, that if we draw a diagram for the factors of 8 we can omit many arrows. It might be tempting to draw something like the diagram on the right.



However, we don't need to draw the arrow $1 \rightarrow 4$ because it's the composite of the arrows $1 \rightarrow 2 \rightarrow 4$; likewise we don't need the arrow $2 \rightarrow 8$. This gives us the following diagram.



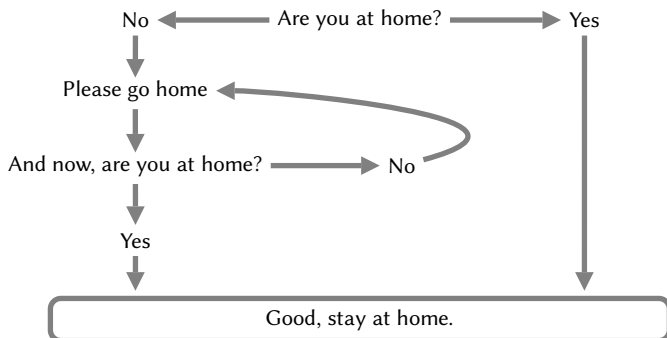
But now we can improve the geometry: there's no reason for the arrows to zig-zag like that, and we might as well draw them in a straight line as shown. I prefer to draw it going up the page vertically rather than sideways, because in all our lattice diagrams the vertical direction on the page has indicated the increasing number of prime factors at each level.



8.8 The point of composition

There is a philosophy that says composition is the whole point of categories. There are, of course, many philosophies about category theory, and we'll mention various of them along the way. The point of this one is to see what the difference is between a category and just a load of arrows.

The following diagram is a COVID lockdown meme, and is a picture of some points and some arrows, like any kind of flow chart. Ostensibly, it looks a bit like a category because it has objects and arrows. But is there anything illuminating to say about composition here?

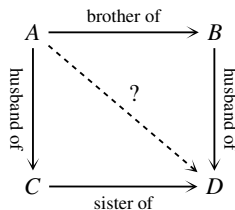


There are several possible paths from “Are you at home?” down to “Good, stay at home”, and you could argue that they’re not the same as each other, but that’s not really the point of the diagram.

Category theory is a lot to do with abstraction and analogies, but the reason composition plays an important role in the definition is that really interesting things happen when composites give rise to interesting information, or give us a way to express interesting information.

When I was growing up we had some family friends who had a situation of family marriages that really intrigued me.

There were two brothers (whom I’ll call A and B) and two sisters (whom I’ll call C and D) and the intriguing thing was that the two brothers married the two sisters. This is perhaps ambiguous linguistically but I hope it’s clear what I mean, and will probably be clearer if I draw a diagram.



One of the things that really intrigued me about this was the relationship between A and D . A was the brother-in-law of D , but in two different ways: he was D ’s husband’s brother, but he was also D ’s sister’s husband. Usually your husband’s brother and your sister’s husband are two different people, but in this case those two relations had converged on each other.

In the language of category theory the two composite arrows from A to D are the same.[†]

Things To Think About

T 8.5 What other relationships can you think of in this family that would end up being the same, where in other families they’re not?

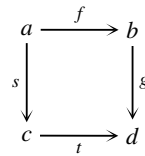
[†] As this is normal life not abstract mathematics there is a slight ambiguity here that I haven’t ironed out. Depending on how we define these arrows we could say that there are two different arrows from A to D and that A is related to D in two different ways. Or not.

Diagrams like this where two composites are (perhaps unexpectedly) the same are called *commutative diagrams* in category theory and are very typical of how we deduce things.

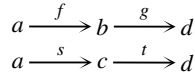
Definition 8.2 A *diagram* in a category is a collection of objects and arrows from the category, possibly not all of them. A diagram is said to *commute* if any two composable strings with the same endpoints produce the same composite.

Note that we refer to diagrams in category theory as shapes just based on how many arrows go around the outside.

So if there are four arrows around the outside we call it a square, regardless of the lengths of the sides and the angles — because those don't affect the logic of the diagram anyway. Commutative squares are the classic building block of category theory.



In that square the only composable strings with the same endpoints are these two paths from a to d .



For the diagram to commute those two composites must be equal, that is

$$g \circ f = t \circ s.$$

If you're very used to writing out equations in algebra like that then you might feel more comfortable with it, but in category theory we don't tend to write those equations, we just draw the diagrams and say they commute. This is because as the diagrams get more and more complex it is often a case of fitting parts together like a jigsaw, and the equations as strings of symbols then become pretty incomprehensible.

When I was thinking about our family friends when I was little I did not think I was doing category theory, but sometimes you're already doing category theory even if you don't realize it, just like you're sometimes doing math even if you don't realize it.

These sorts of real life category theory situations are admittedly not exactly rigorous, and they're not really profound category theory either. However, they are a way in which I can clarify my thinking about the world around me by drawing on the techniques of category theory and the way of thinking that it provides. In the next part of the book we'll take an interlude and delve into some more rigorously mathematical examples.

INTERLUDE

A TOUR OF MATH

